

Probability - I
Midterm Examination
06/10/2025

1. An electronics store has received a shipment of 14 table radios that have connections for an iPod or iPhone. Eight of these radios have two slots so that they can simultaneously accommodate two devices, and the other six have a single slot. Suppose that five of the fourteen radios are randomly selected to be stored under a shelf where the radios are displayed, and the remaining ones are placed in a storeroom. What is the probability that there are at least two radios of each type among the displayed radios? [4]
2. A large operator of timeshare complexes requires anyone interested in making a purchase to first visit the site of interest. Historical data indicates that 20% of all potential purchasers select a day visit, 50% choose a one-night visit and 30% opt for a two-night visit. In addition, 10% of day visitors ultimately make a purchase, 30% of one-night visitors buy a unit, and 20% of those visiting for two nights decide to buy. Suppose a visitor is randomly selected and is found to have not made the purchase. How likely is it that this person made a one-night visit to the site? [5]
3. Let $X \sim \text{Poi}(\lambda)$ where $\lambda > 0$. Find $E(X)$, $E(X^2)$ and $E(X^3)$. [6]
4. A library subscribes to two different weekly news magazines, each of which is supposed to arrive in Wednesday's mail. In actuality each one may arrive on Wednesday, Thursday or Friday. Suppose the two arrive independently of one another and for each one, $P(\text{Wed.}) = .4$, $P(\text{Thurs.}) = .3$, $P(\text{Fri.}) = .3$.
Let Y = the number of days beyond Wednesday that it takes for both magazines to arrive. (e.g. If both the magazines arrive on Wednesday, then $Y = 0$ and if one arrives on Thursday and the other on Friday, then $Y = 2$.) Compute the pmf of Y , $E(Y)$ and $V(Y)$. [6]
5. Let Ω be the set of Roman alphabets. i.e. $\Omega = \{a, b, c, \dots, x, y, z\}$. Consider four events $A_1 = \{a, b, c\}$, $A_2 = \{b, c, d\}$, $A_3 = \{a, c, d\}$ and $A_4 = \{a, b, d\}$. Suppose P is a probability measure such that $P(A_i) = \alpha_i$ for $i = 1, 2, 3, 4$. Find $P(\cup_{i=1}^4 A_i)$. [4]